



Your team's memory, on every laptop. Even with no signal.

Offline-first team memory on Qdrant Edge. When two people disagree about a deadline, it asks instead of guessing.

Code Cubicle 6.0 · Problem Statement 03 · by BKA

same job, dates disagree

Client just called:
Sharma delivery
moves to the
18th.

note · Tanishk

Confirming final
delivery of the
Sharma wedding
edit on 16
October.

email · Lakshya

Ask us about

The problem

why field teams lose track

Your private assistant

what it knows, and why it can

Catching conflicts

the Qdrant query

Tech stack

and why each piece

FAQ: limits

scale, clocks, dates

What Quorum does

four things, in one line each

A day with Quorum

the Sharma day, across devices

Sync and offline

outbox, cursor, switch

Real vs mocked

straight answers

What's next

after this round

How it works

interactive architecture

A claim

memory with a source

Privacy

what never leaves

FAQ: design

key vector, LLM, 1.00

deep dive [architecture \(interactive\)](#) [module map](#) [capture pipeline](#) [search](#) [conflict engine](#) [sync engine](#) [data model](#) [failure handling](#) [trade-offs](#)

A deadline changes in one place and gets misquoted in another.

Many clients, many channels. Four people each hear a different piece.

email · Lakshya

Confirming final delivery of the Sharma wedding edit on 16 October.

Client just called: Sharma delivery moves to the 18th.

a call on set · Tanishk

Sharma delivery Monday?

WhatsApp

~~!!!~~ **Often no signal**

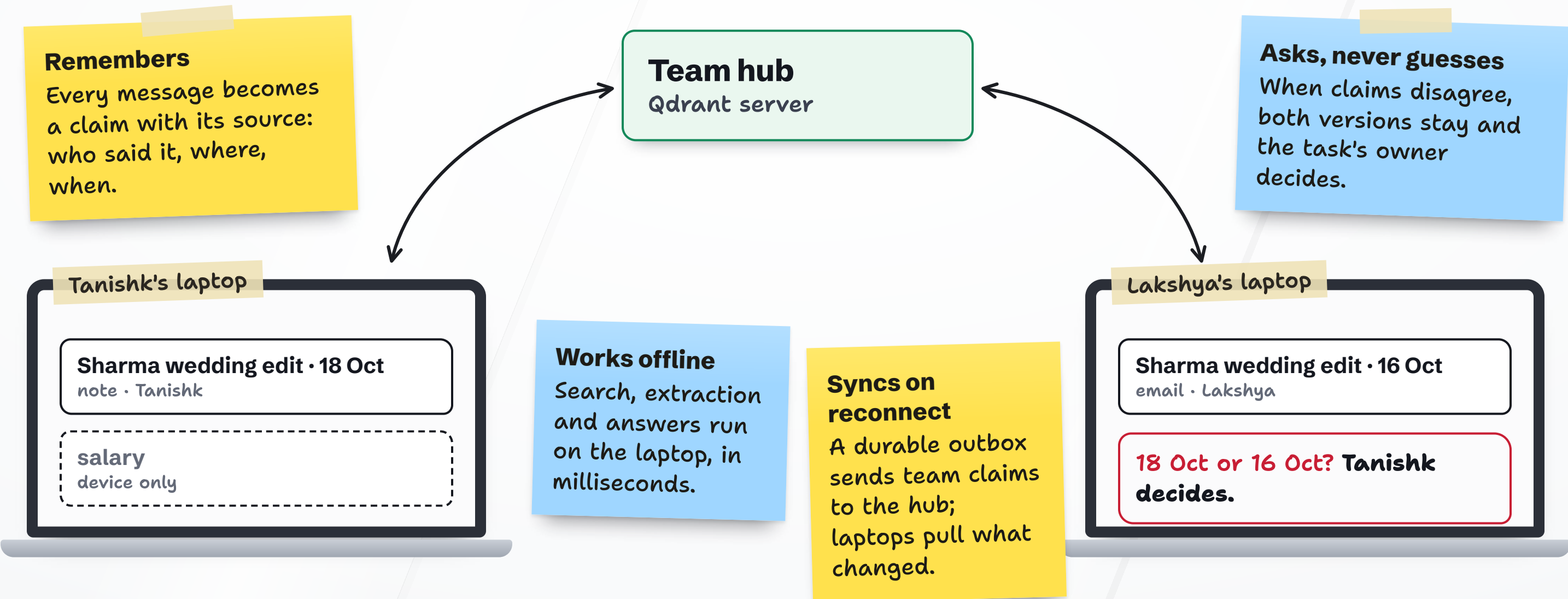
Shoots and site work happen where cloud tools don't reach.

Cloud assistants don't fit

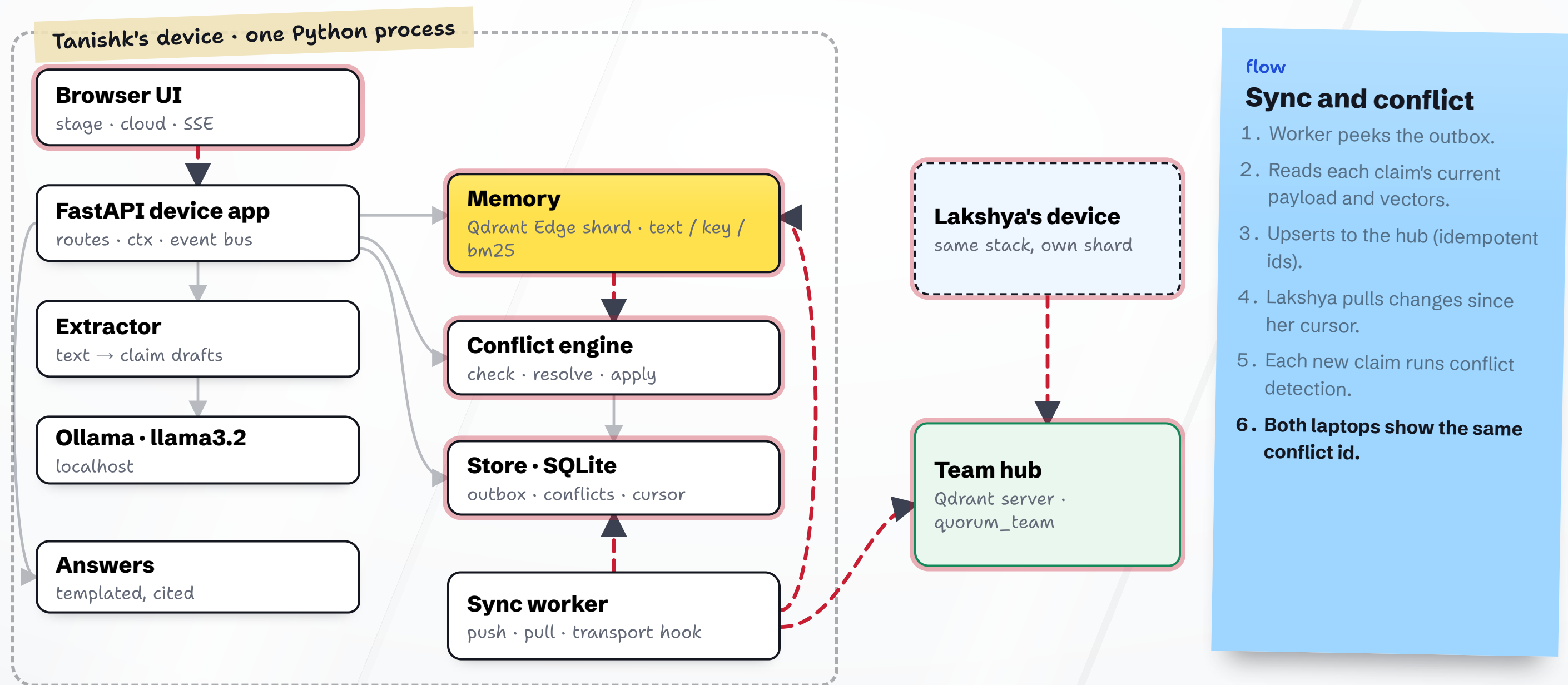
To help, they'd need everything: your inbox, your chats. That shouldn't leave the laptop.

Nobody notices until the client does.

One memory for the team. Every laptop keeps its own copy.



How the system fits together.



For this round both laptops run on one machine as separate processes, each with its own shard. "Offline" is a per-device switch that cuts hub traffic at the transport; everything else keeps running.

Your private assistant.

it reads

email

WhatsApp

calls and notes

what you're working on
active app and window

the phone in your pocket

your laptop

Quorum

knows what you're up to,
because it never leaves the
laptop

—||| voice in and out

hyperfocus nudge

"You've been on the
game project since 2.
The Sharma edit is due
tomorrow at 10."

deadline reminder

"Reel cutdowns due
Monday. Three, not
two, per the client's
last message."

a person

"Riya's birthday is
Thursday. Last time
you talked about her
trip to Goa."

a pattern

"You usually forget
invoices on Fridays.
Want a reminder at
4?"

It can know this much because nothing leaves your laptop unless it's work your team needs.

A day with Quorum.

Tanishk at BKA, a four-person production house and developer group

morning · laptop

the shoot · phone

afternoon · laptop

reconnect · both
laptops

evening · everyone

no signal, still working

A brief from his own memory: today's shoot, and what's due this week.

No signal in Ladakh. The client calls: delivery moves to the 18th. A voice note on the phone, saved on the device.

"You've been on the game project since 2. The Sharma edit is due tomorrow at 10."

Back in signal: his note goes up, Lakshya's email comes down. Both laptops catch 18 Oct vs 16 Oct and ask Tanishk, the owner.

Tanishk keeps the 18th. Everyone has it a second later. His private notes stayed private.

Memory is claims, not facts.

Client just called:
Sharma delivery
moves to the 18th.

note · Tanishk · 3 Oct, 14:02



claim	
entity	Sharma wedding edit
attribute	due date
value	18 Oct
source	call, heard by Tanishk
tier	team
status	active · disputed · superseded

three vectors per claim

text

meaning

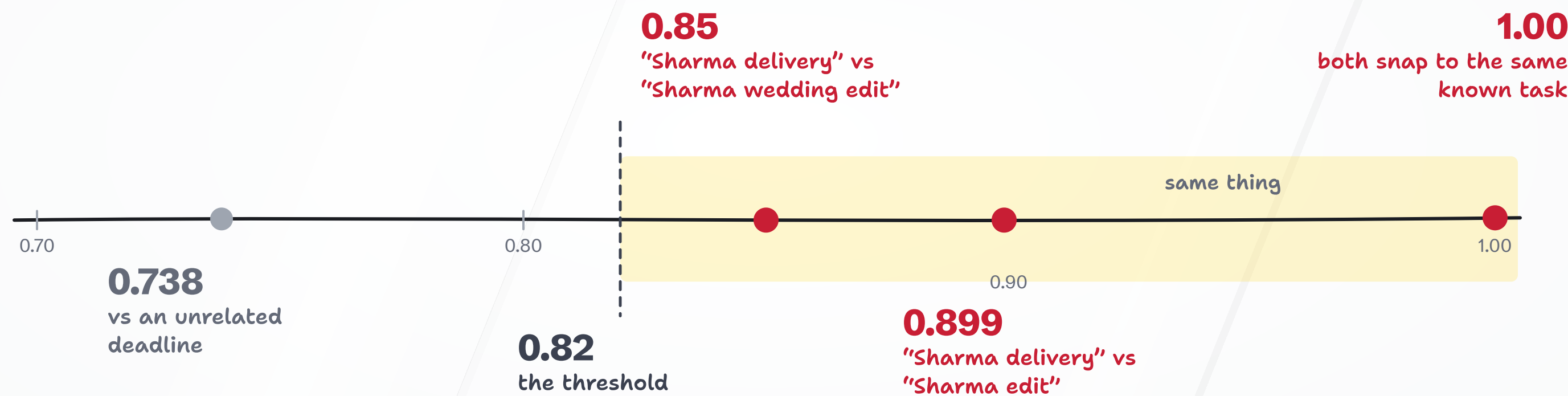
bm25

keywords

key

what it's about
entity + attribute,
never the value

Qdrant finds claims about the same thing.



Every new claim, typed or pulled from the hub, runs one query on the **key** vector.

Filtered to the same attribute and live claims; threshold 0.82.

Same thing, different value: a conflict. Nothing is overwritten.

Both laptops agree without talking: the id is a hash of the two claim ids.

The task's owner keeps one; the other stays as history; the decision syncs.

Works where the signal doesn't.

offline switch

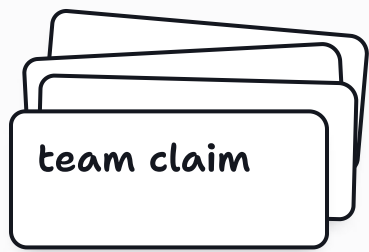
one transport hook refuses hub traffic and counts bytes, so **0 B** is measured, not drawn

Team hub
Qdrant server



pull

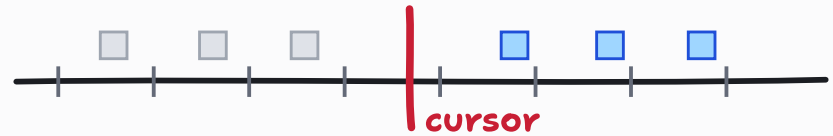
Tanishk's laptop



- team claims queue
- survive restarts
- push oldest first
- retry with backoff

outbox • SQLite

Lakshya's laptop



- scrolls the hub for claims changed since its last pull
- points sync with vectors, no re-embedding

7-70 ms

search on the laptop

~1 s

conflict on both laptops after reconnect

~1 s

a decision reaches the other laptop

It reads everything. Nothing leaves unless it's team work.

salary · a friend's news

Device only
never leaves

birthdays · my schedule

My devices
my laptops only

deadlines

shoots

deliverables

Team
everyone

Team hub

0 private claims on the hub,
counted live on stage

- Everything starts device-only; rules promote only clear team work.
- A follow-up never gets a wider tier than its entity already has.
- The outbox and the push both refuse device-only claims.
- The pull only takes team claims (or my own).
- Team view filters by tier inside the Qdrant query.

What we used, and why.

Qdrant Edge

Vector DB on the laptop:
BM25 + dense + filters in one
query, offline. Same data
model as the hub.

Qdrant server

The team hub. Points
sync as-is, vectors
included.

bge-small (FastEmbed)

Small, fast, local
dense embeddings.
384 dims.

BM25 + RRF

Exact names
("Sharma") plus
meaning; RRF fuses by
rank, no score
calibration.

Ollama • llama3.2

Local extraction, no cloud
calls. Dates are resolved in
code, not by the model.

SQLite

Durable outbox,
conflicts and sync
cursor.

FastAPI + vanilla JS

Two device processes and a
live UI over SSE. No build
step.

Straight answers.

Real

- ✓ Qdrant Edge memory and hybrid search on each device
- ✓ Conflict detection at capture and at sync
- ✓ Hub sync with a persistent outbox
- ✓ Owner resolution that syncs
- ✓ Tier privacy, zero private claims on the hub
- ✓ Local extraction with sources

Mocked or next

- Two physical laptops (one machine this round)
- Gmail API (a fixture inbox stands in)
- Phone app, live WhatsApp stream
- Partial snapshots, encryption at rest
- Hub auth and TLS

FAQ: design

Why a separate key vector?

A conflict is "same thing, different value", so matching must ignore the value and the wording. The key is just entity + attribute.

Why not let the LLM decide conflicts?

Two laptops must reach the same conflict on their own: same vectors, same query, same threshold, same id. The LLM only extracts.

Why does the demo show similarity 1.00?

Both mentions were matched to the known task "Sharma wedding edit", so the key text is identical. Worded differently, it's 0.85–0.90.

Why templated answers?

Milliseconds, no hallucinated facts, and every sentence cites a claim.

FAQ: limits

What if the model misreads a date?

It never computes dates; code resolves them from the message. And a wrong claim can't overwrite anything: it shows up as a conflict with both sources.

Does it scale?

Each question is one hybrid query or one filtered nearest-neighbour search, which is what Qdrant is built for. We haven't benchmarked large memories yet.

**Three devices?
Clock skew?**

Detection scales: each device finds the same conflict. The sync cursor uses device clocks; the fix is a hub-assigned sync stamp.

Business model?

Per team, per month; no price set yet. Customer zero is BKA, a four-person production house and developer group.

After this round.

Two laptops

Real airplane mode, a hub-assigned sync stamp, partial snapshots.

Hardening

Hub auth and TLS, encryption at rest, N-way conflicts, packaging.

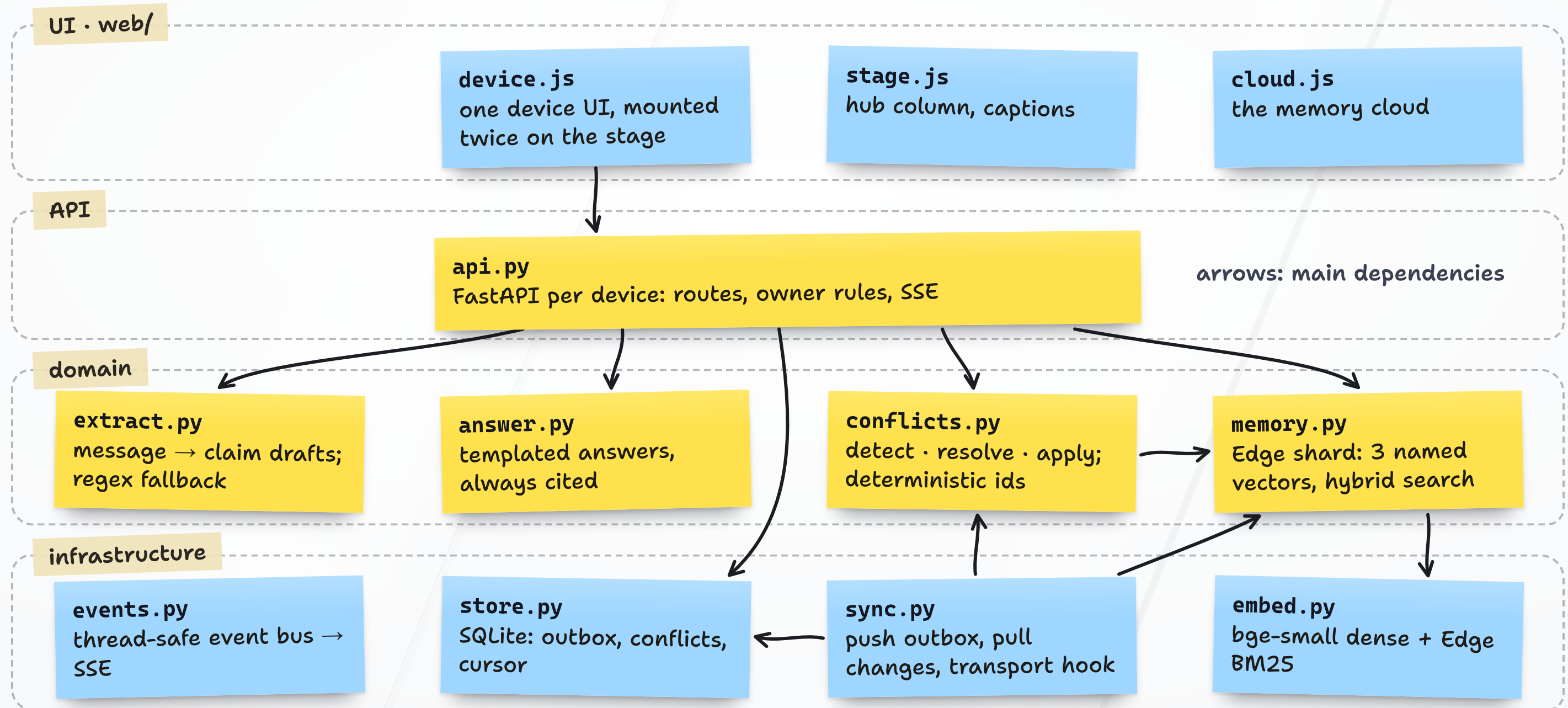
More inputs

Gmail API, WhatsApp import in the UI, phone capture.

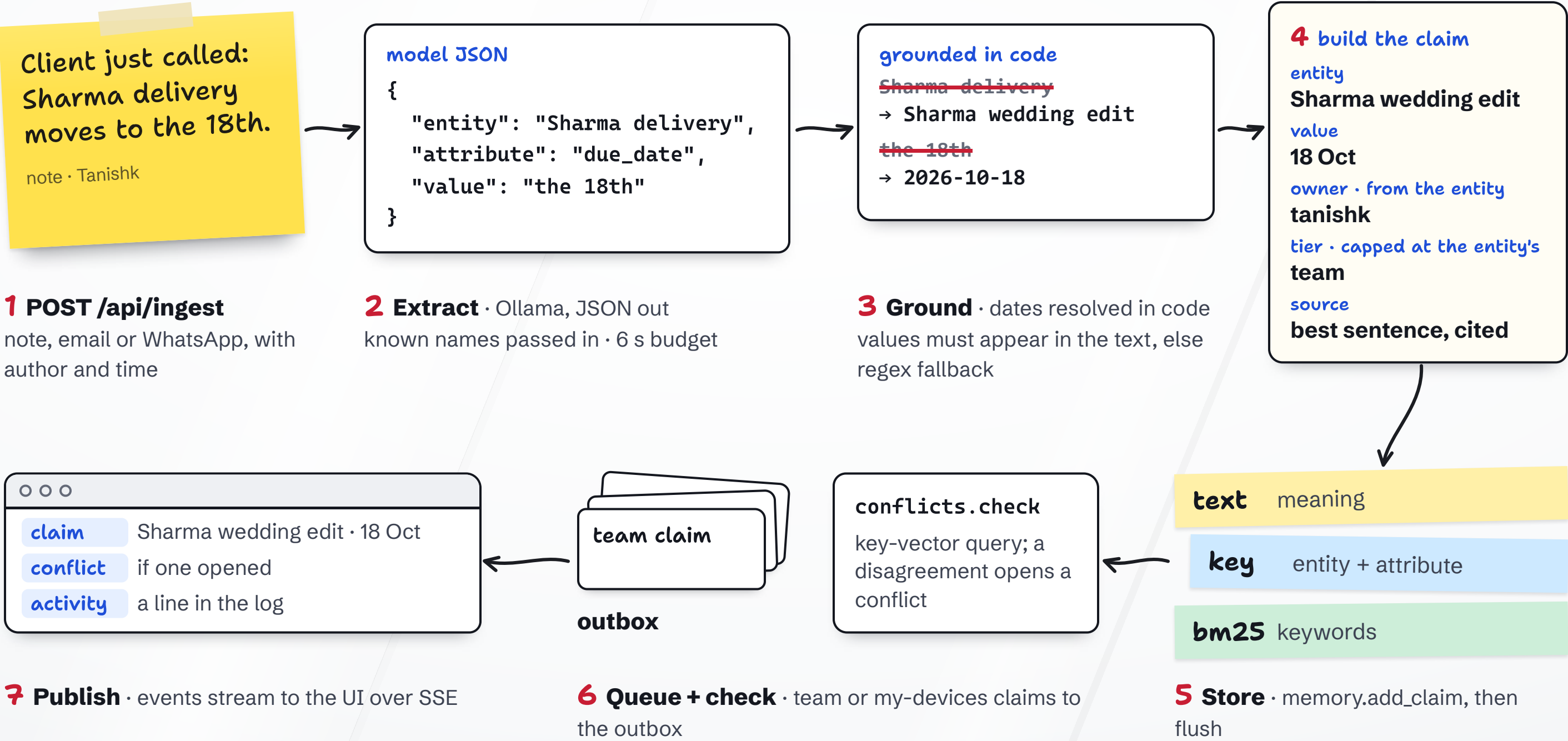
this round

Thank you. Ask us anything.

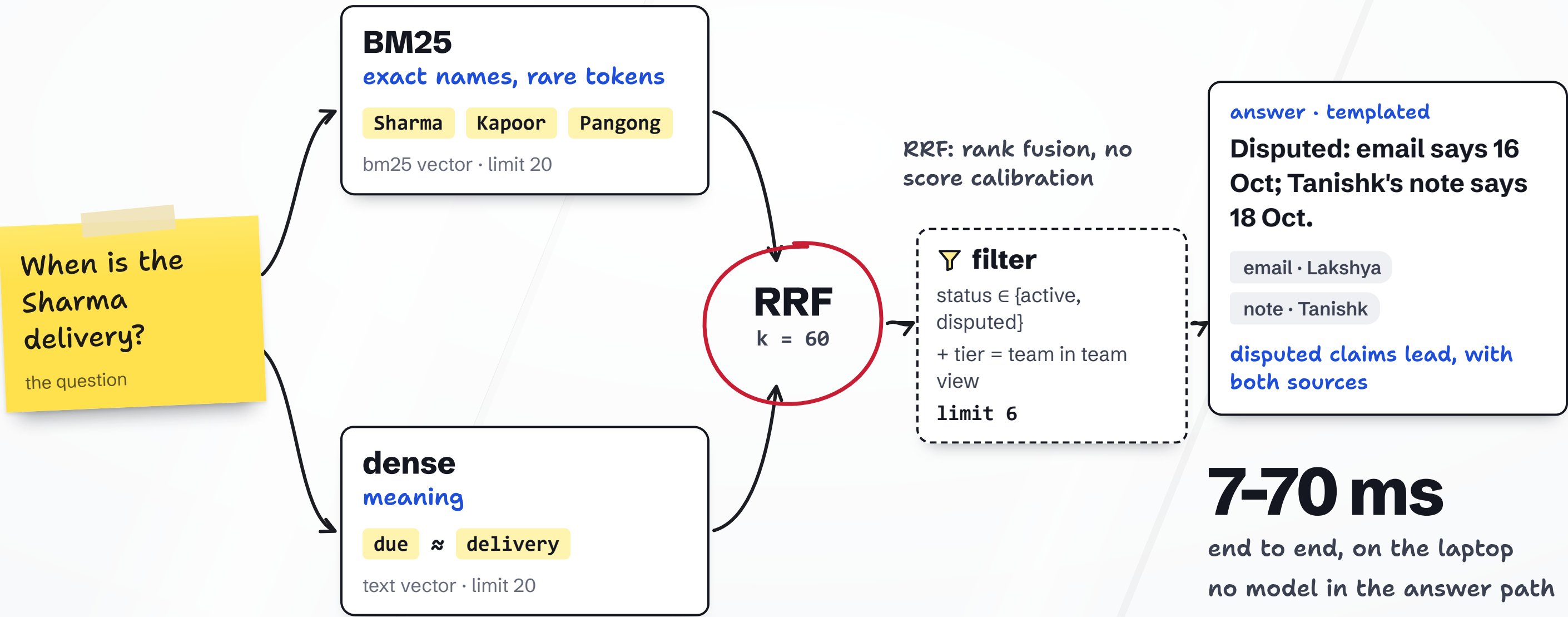
Module map: one job per file.



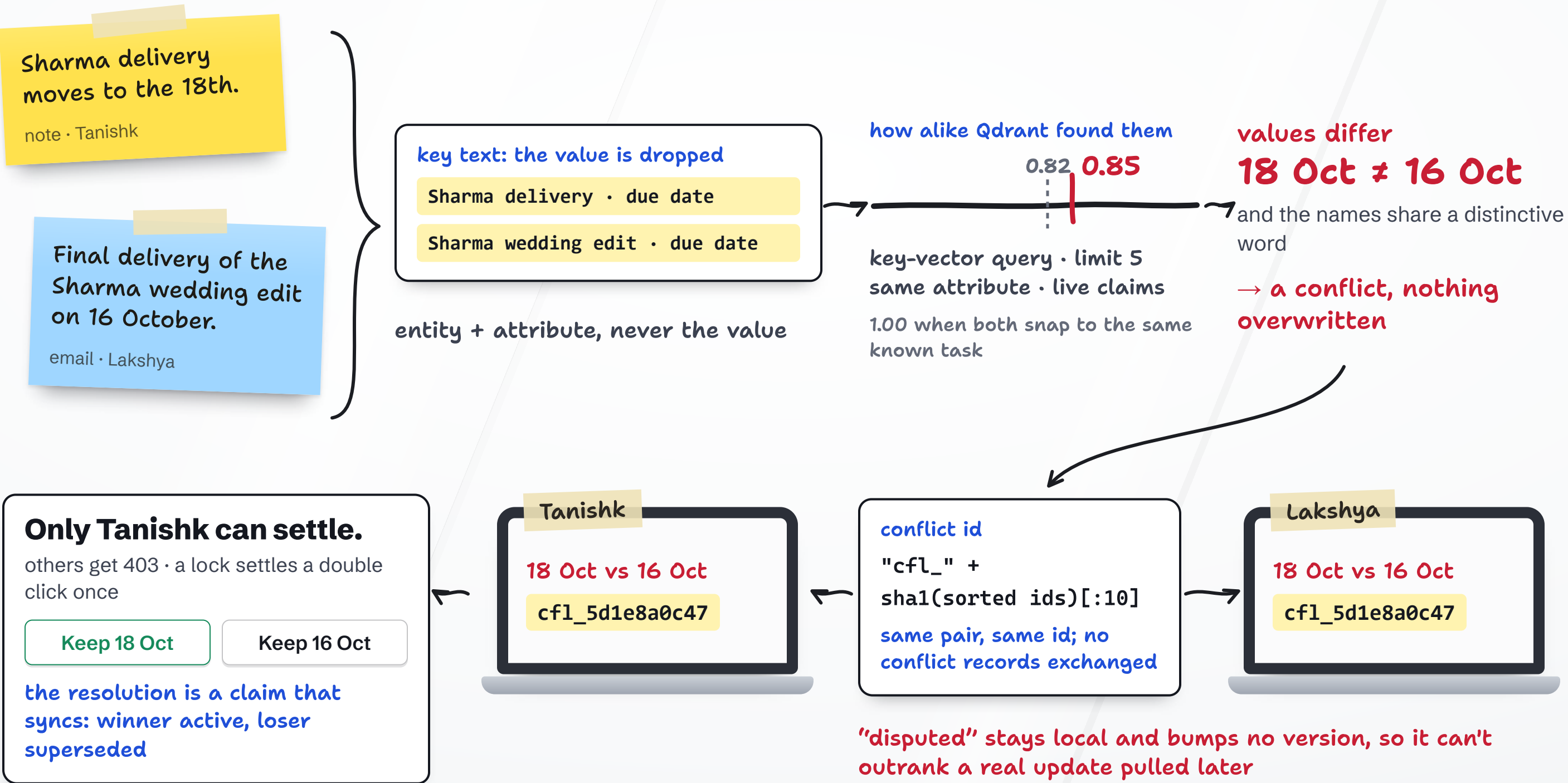
Capture: from a message to a claim.



Search and answers: one query, on the laptop.



Conflict engine: deterministic on every laptop.



Sync engine: outbox up, cursor down.

offline switch

the transport hook refuses every request and counts bytes

Team hub

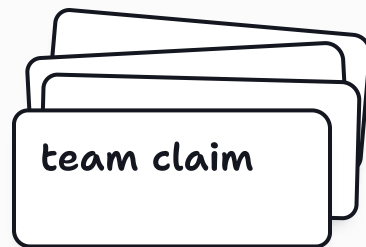
Qdrant server · quorum_team
a relay: no logic of its own



push · every second

pull · scroll
the hub

Tanishk's laptop



outbox · SQLite

- oldest change first
- batches of 64
- uuid5 ids: retries are idempotent
- acks check a gen counter
- device-only dropped

Hub down?

Backoff up to 5 s. The outbox keeps everything.

Lakshya's laptop



- not modified by me; team, or my own my-devices
- new claim → stored with its vectors, then conflict check
- known claim → applied only if its version is newer

Data model: a point per claim.

claim

point id = uuid5(claim_id)

claim_id	"clm_3f9a1c2b7d4e"
entity	"Sharma wedding edit"
attribute	"due_date"
value	"2026-10-18" · value_label "18 Oct"
owner	"tanishk"
source	{kind, ref, author, excerpt, at}
tier	device my_devices team
status	active disputed superseded
version · modified_at · modified_by	
conflict_id · resolves	

text

key

bm25

three named vectors

dense 384 · cosine
the claim sentence

dense 384 · cosine
entity + attribute

sparse · IDF
the claim sentence

- modified_at
- tier
- modified_by
- captured_by
- attribute

quorum_team
hub collection: the same named vectors
payload indexes (the tabs) serve the pull filter

Concurrency and failure handling.

Crash mid-write?

Every write flushes (~28 ms), so a hard kill loses nothing. SQLite runs in WAL mode; the outbox survives restarts.

Two threads at once?

API and sync threads share one Edge shard behind a re-entrant lock. SQLite: one connection, serialised by a lock. One sync round at a time; a pull applies in one batched flush and the cursor is saved only after it lands.

Hub down?

Backoff to 5 s, "Hub down" in the UI. The collection is re-created if the hub came back empty.

Model slow or down?

6 s budget per message. After a connection error the model is skipped for 20 s; the regex fallback runs.

SSE drops?

15 s heartbeat; the page reloads state and reconnects.

Double click on Keep?

A lock: it settles once. Only the owner may resolve (403 otherwise).

Known limits

- The pull cursor uses device clocks
fix: a hub-assigned stamp
- Conflicts are pairwise
- The hub has no auth yet

Decisions and trade-offs.



A key vector

over
~~entity ids~~

Same thing in different words still matches;
no global id service.

cost: one extra vector per claim



A deterministic rule decides conflicts

over
~~the LLM~~

Both laptops agree offline; the similarity is measurable.

cost: the threshold must be tuned



Templated answers

over
~~generated prose~~

Milliseconds, cited, no hallucination.

cost: less fluent prose



Hub as relay only

over
~~logic on the hub~~

All intelligence stays on devices; the hub can be any Qdrant server.

cost: needs a reachable server to share



Scroll by cursor

over
~~partial snapshots~~

Simple, exact for two devices.

cost: device-clock cursor; snapshots next



Python bindings + FastAPI

over
~~Tauri / Rust~~

Built and verified in hours; the same Edge shard underneath.

cost: not a single packaged binary yet

Also: two processes on one machine, not two laptops. One presenter, one shared screen this round; airplane mode is shown as a hub switch.